

Peripheral Nerve Sheath Tumors

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Learning Objectives (6)

After completing this brick, you will be able to:

- 1 Define a peripheral nerve sheath tumor, and list the common benign and malignant types.
- 2 Describe a schwannoma and its histologic features and clinical presentation.
- 3 Explain how the *NF2* gene mutation leads to proliferation of Schwann cells.
- 4 Describe the histologic features of neurofibroma, list the three clinical variants, and explain which cells contain a mutation in the *NF1* gene.
- 5 Describe malignant peripheral nerve sheath tumors.
- 6 Describe, compare, and contrast the incidence, clinical features, and genetic basis of neurofibromatosis type 1 and type 2.

CASE CONNECTION

EB is a 64-year-old male who comes to your office with ringing in his left ear. “It’s downright maddening, especially when I try to go to sleep. It’s a high-pitched, irritating humming,” he says. “I’m also having some trouble hearing, mostly in my left ear. And if that’s not enough, sometimes when I move quickly, I feel like I’m spinning, like the sensation you get when you come off a rollercoaster. I don’t know if all these things are related, but they all have occurred over the past year.”

What is the likely diagnosis? What treatment will you recommend for EB? Consider your answers as you read, and we’ll revisit EB at the end of the brick.

[GO TO CONCLUSION](#) ↓

What Are Peripheral Nerve Sheath Tumors?

When tumors develop along the nerves of the peripheral nervous system, they are called peripheral nerve sheath tumors. Overall, these tumors are uncommon. But two conditions, known as neurofibromatosis type 1 and type 2, predispose patients to developing many peripheral nerve sheath tumors.

INSTRUCTOR NOTE

See handout posted in Canvas

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Some of these tumors, like schwannomas and neurofibromas, can be benign. They can also be malignant—most commonly occurring in neurofibromatosis type 1. We start this discussion by looking at specific types of peripheral nerve sheath tumors, and then we look at neurofibromatosis types 1 and 2.

What Are Schwannomas?

Schwannomas are the most common nerve sheath tumors. They are often found incidentally when neuroimaging studies are performed. In fact, 3%-4% of people demonstrate cranial nerve schwannomas on autopsy.

How Schwannomas Present

Schwann cells are responsible for myelination of nerves in the peripheral nervous system. Schwannomas are benign, fully encapsulated tumors composed of neoplastic Schwann cells.

Schwannomas can be found within soft tissues, on cranial nerves, surrounding peripheral nerves, and within internal organs. They are typically small, slow-growing masses that present with effects of compression of local structures or as incidental findings on imaging.

INSTRUCTOR NOTE

Schwannomas can usually be removed from the peripheral nerve it is attached to. This is in contrast to plexiform neurofibromas (see below) which infiltrate into the nerve.

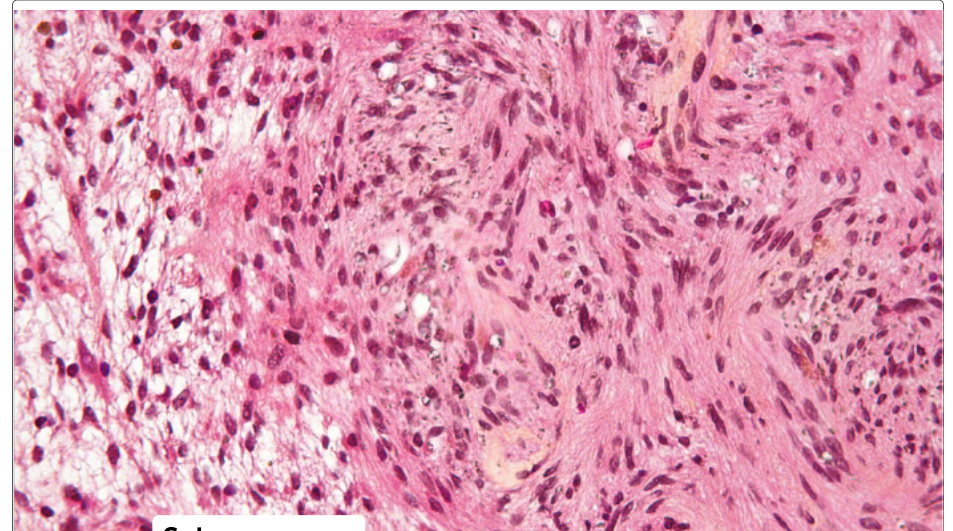
below) which infiltrate into the nerve.

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There is often no need to treat schwannomas. If the tumor is symptomatic or in a position to compromise nearby structures, modalities include surgical excision or, if intracranial, stereotactic radiosurgery.

What Schwannomas Look Like

Grossly, schwannomas are well-circumscribed masses attached to an adjacent nerve. Although schwannomas are homogeneously composed of Schwann cells, on microscopic evaluation they demonstrate characteristic dense and loose areas referred to as **Antoni A and B**, respectively (Figure 1).





QUIZ

Tap image for quiz

Figure 1

Within the densely packed Antoni A, diagnostic **Verocay** bodies may be observed (Figure 2). Verocay bodies demonstrate cells that align to produce nuclear palisading—alternating nuclear and anuclear areas.

Figure 2

What are the most common nerve sheath tumors?

Causes of Schwannoma Formation

Schwannomas, both sporadic and familial, form because of loss of function of a protein called **merlin**. Merlin is encoded by the *NF2* gene on chromosome 22. This cytoskeletal protein functions as a tumor suppressor through E-cadherin–mediated contact inhibition of proliferation. Contact inhibition of proliferation is a process that prevents overgrowth of tissues by signaling cells to discontinue proliferation when they are densely surrounded by neighboring cells.

With loss of function of the merlin protein, Schwann cells proliferate unchecked by contact inhibition. As is traditionally understood with tumor suppressor genes, both copies of the *NF2* gene must be nonfunctioning to result in tumor growth, which is more likely to occur in patients already missing one copy. This is the case in patients with neurofibromatosis type 2 (NF2), a familial condition we discuss in greater detail later.

The function of what protein is implicated in the development of schwannomas?

Vestibular Schwannoma

Schwannomas often affect the cranial nerves, most commonly the vestibular portion of the eighth cranial nerve. Historically, schwannomas on the eighth cranial nerve were referred to as acoustic neuromas. Because these tumors are composed of neoplastic Schwann cells on the vestibular nerve, they are more accurately called vestibular schwannomas, but you will see these terms used interchangeably. (Some neuropathologists will be unhappy if you call this tumor anything but a vestibular schwannoma because the tumor cells are not neurons, which makes acoustic neuroma incorrect.)

Vestibular schwannomas arise near the brainstem or within the internal acoustic canal. Patients usually present with unilateral hearing loss, ear fullness, tinnitus, or vestibular complaints such as dizziness.

When lesions occur at the cerebellar pontine angle at the origination of the eighth cranial nerve (CN), we would expect to see effects at the nearby seventh cranial nerve. However, patients with lesions in this area more commonly present with issues involving the trigeminal nerve (CN V) because the facial nerve (CN VII) is less easily compressed. As the tumor enlarges, it may compress the ipsilateral cerebellum and cause ipsilateral ataxia (abnormal, uncoordinated movement).

Most vestibular schwannomas occur spontaneously and are unilateral. However, about 10% of tumors are found in patients with neurofibromatosis type 2. These patients are also much more likely to have bilateral tumors, which are pathognomonic for this condition. [Figure 3](#) is a magnetic resonance image (MRI) showing bilateral masses in the cerebellopontine angles with extension into the internal acoustic meatuses in a patient with NF2.





QUIZ

 Tap image for quiz

Figure 3

CLINICAL CORRELATION

Depending on the size and location of a vestibular schwannoma, a patient may elect to manage the disease with microsurgery, radiotherapy, or conservative management. Microsurgery is performed by a specialized neurosurgeon and is used for larger tumors. Gamma knife is a form of stereotactic radiosurgery performed by a radiation oncologist in collaboration with neurosurgery. Gamma knife involves a single treatment of high-dose radiation to the tumor, which is intended to prevent future growth. Conservative management usually involves serial computed tomographic (CT) scans with treatment after the mass exceeds certain growth parameters.

What Are Neurofibromas?

Like schwannomas, neurofibromas are benign, solitary lesions that are commonly sporadic. However, about 10% of neurofibromas arise in

patients with neurofibromatosis type 1 (NF1).

How Neurofibromas Present

Although neurofibromas and schwannomas are both benign, neurofibromas—unlike schwannomas—can occasionally demonstrate transformation into malignant peripheral nerve sheath tumors. Neurofibromas are typically seen in patients younger than 30 years; they can appear as small nodular tumors but may also be large and disfiguring. Most commonly tumors present as skin-colored or hyperpigmented small papules or nodules. Lesions are usually asymptomatic in terms of focal neurologic deficits, but patients may describe bleeding or local pain from a mass effect.

What Neurofibromas Look Like

Neurofibromas are similar to schwannomas, benign tumors comprising neoplastic Schwann cells. However, unlike schwannomas, neurofibromas are not encapsulated, and they contain other cell types such as mast cells, fibroblasts, and perineurial-like cells mixed among the neoplastic Schwann cells. [Figure 4](#) shows a neurofibroma that demonstrates loose bands of spindle-shaped cells when viewed under light microscopy.

INSTRUCTOR NOTE

Another term for this look is "shredded carrots"

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Figure 4

Subtypes of Neurofibromas

There are three distinct subtypes of neurofibromas: localized cutaneous, plexiform, and diffuse.

Localized cutaneous neurofibromas (Figure 5) often occur in sporadic cases, but multiple lesions may arise in the context of NF1. These lesions are circumscribed, nodular, or polypoid tumors.

QUIZ

Tap image for quiz

Figure 5

Plexiform neurofibromas grow within the confines of a nerve or nerve plexus. They involve multiple fascicles of individually affected nerves. Axons may be embedded within a diffuse neoplastic Schwann cell proliferation. This proliferation causes nerve fascicles to expand, but the perineurium remains intact. These tumors are highly associated with NF1 and carry a small risk of malignant transformation. The tumors can be disfiguring (Figure 6).

Figure 6

Diffuse neurofibromas demonstrate extensive infiltration within the dermis or subcutis of the skin and are often associated with NF1. They can present as large, disfiguring subcutaneous masses.

How are neurofibromas histologically different from

How are neurofibromas histologically different from schwannomas?

Causes of Neurofibroma Formation

Neurofibromas arise from nonmyelinating Schwann cells that demonstrate loss of function of the tumor suppressor protein **neurofibromin** on chromosome 17. Neurofibromin acts as a negative regulator of the Ras oncoprotein. Ras activity drives cellular proliferation. Ras becomes hyperactive when inhibition by neurofibromin is removed and leads to unchecked cellular proliferation.

Patients with neurofibromatosis type 1 lack one copy of neurofibromin, which predisposes them to the development of neurofibromas. We look at the familial condition NF1 in greater detail below.

What Are Malignant Peripheral Nerve Sheath Tumors?

Malignant peripheral nerve sheath tumors (MPNSTs) are rare but represent a very serious form of nerve sheath pathology. Up to about 50% of MPNSTs occur in the context of neurofibromatosis type 1. When treating these tumors, you need to consider surgery, medical oncology, and radiation oncology.

These tumors most commonly present on the trunk or extremities and are

These tumors most commonly present on the trunk or extremities and are usually discovered because of rapid tumor growth. On gross inspection, tumors are large and firm and can demonstrate areas of necrosis and hemorrhage. Microscopic evaluation demonstrates densely cellular, mitotically active collections of spindle cells.

MPNSTs may grow spontaneously from previously normal nerves or result from malignant transformation of a plexiform neurofibroma. About 10% of cases occur as a late complication of radiation therapy.

Neurofibromatosis

We mentioned NF1 and NF2 in the context of different tumor types. Now let's look at them in more detail.

Neurofibromatosis type 1, also known as von Recklinghausen disease, predisposes patients to developing neurofibromas through loss of function of neurofibromin. This protein is encoded by a gene found on chromosome 17. NF1 has an incidence of 1 in 3000 people, and patients are at increased risk of developing pheochromocytomas, similar to multiple endocrine neoplasia 2 (MEN2).

Neurofibromatosis type 2 predisposes patients to developing schwannomas through loss of function of merlin. This protein is encoded by a gene on chromosome 22. NF2 has an incidence of about 1 in 25,000, making it much less common than NF1.

INSTRUCTOR NOTE

Robbins cites a different incidence - see power point (1/40,000 - 50,000) - but the point is that NF2 is less common than NF1

50,000) - but the point is that NF2 is less common than NF1

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“Neurofibromatosis” has **17** letters; NF1 is on chromosome **17**.
“Neurofibromatosis type **2**” has **22** characters (without spaces);
NF**2** is on chromosome **22**.

How Neurofibromatosis Presents

Patients with NF1 may have **Lisch nodules**, which are hamartomas of the iris that do not affect vision (**Figure 7**). Lisch nodules are the most specific feature of NF1.



QUIZ

 Tap image for quiz

Figure 7

Patients with NF1 demonstrate significant **freckling**, especially in an axillary and inguinal distribution, as well as hyperpigmented macules and patches known as **café au lait spots** (French for coffee with milk, referring to the color). These spots are usually evident within the first year of life (Figure 8).



QUIZ

 Tap image for quiz

Figure 8

Patients with NF1 are predisposed to all subtypes of neurofibromas. Optic pathway gliomas occur in up to 15% of children with NF1, typically before 6 years of age, so these children need regular formal ophthalmology evaluations. An MRI of the brain and orbits is indicated if the child has any visual disturbances.

Roughly 10% of patients with NF1 develop a malignant peripheral nerve sheath tumor at some point in their lifetime. Patients with NF1 also may exhibit learning disabilities, seizures, vascular abnormalities, and skeletal

abnormalities.

Rather than neurofibromas, patients with NF2 are predisposed to develop schwannomas. NF2 can present with bilateral vestibular schwannomas, other cranial nerve schwannomas, cataracts, meningiomas, and cutaneous tumors.

Table 1 summarizes the differences between NF1 and NF2.

Table 1

	NF1	NF2
Inheritance	Autosomal dominant	Autosomal dominant
Chromosome	17	22
Protein product	Neurofibromin	Merlin (tumor suppressor)
Clinical features	Hyperpigmented macules, axillary or inguinal freckling, Lisch nodules (iris hamartomas), neurofibromas, optic gliomas	Bilateral vestibular schwannomas, schwannomas of other cranial nerves, intracranial meningiomas, peripheral neuropathy, cataracts, cutaneous tumors

Treatment and Prognosis for NF1 and NF2

There is no definitive overall treatment for NF1, but patients should be screened frequently for other problems commonly associated with the disease. Patients typically require full body skin surveys, ophthalmologic examinations, academic support, and monitoring for hypertension, bone abnormalities, and seizures. There are limited mortality data, but the lifespan for patients with NF1 is about 15 years shorter than that for the average population.

Patients with NF2 require surveillance for tumors. Schwannomas associated with NF2 have responded to bevacizumab (a monoclonal antibody therapy), but radiation therapy and surgery are frequently required. Although radiation is appropriate for treating tumors, it should be minimized in these patients because of the likelihood of secondary tumors. On average, patients with NF2 present at around age 20 years and die in their 30s due to complications from their tumors.



CASE CONNECTION

[BACK TO INTRODUCTION](#) ↑

Thinking back to EB, what is the likely diagnosis? What treatment do you recommend?

The triad of unilateral hearing loss, tinnitus, and vertigo is consistent with a vestibular schwannoma, a benign tumor composed of neoplastic Schwann cells. You order a CT scan, which reveals a large tumor in the left internal

What are the common dermatologic findings in patients with neurofibromatosis type 1?

cells. You order a CT scan, which reveals a large tumor in the left internal auditory canal affecting the vestibular portion of CN VIII. In consultation with a neurosurgeon, EB elects to have surgical resection. At his 3-month follow-up visit, his hearing loss persists, but his tinnitus is markedly improved. "I can live with the hearing loss, but that ringing in the ears was a killer. I'm so glad it's gone," EB says.

Summary

- Schwannomas are benign, encapsulated proliferations of neoplastic Schwann cells.
- Microscopically, schwannomas demonstrate dense and loose areas known as Antoni A and B, respectively.
- Verocay bodies are characteristic features of schwannomas, exhibiting palisading nuclear and anuclear regions.
- Sporadic and familial schwannomas arise because of loss of function of the merlin protein.
- Neurofibromas are benign proliferations of Schwann cells that also contain other cell types.
- The three distinct subtypes of neurofibromas are localized cutaneous, plexiform, and diffuse.
- Neurofibromas form because of loss of function of the neurofibromin protein.
- Neurofibromin acts as a negative regulator of the Ras cell growth pathway.
- Malignant peripheral nerve sheath tumors (MPNSTs) are rare but serious cancers, and patients with NF1 are particularly at risk; MPNSTs may arise from normal nerves or existing plexiform neurofibromas.
- NF1 presents with Lisch nodules, inguinal and axillary freckling, café au lait spots, optic gliomas, neurofibromas, and other abnormalities.
- NF2 presents with cranial nerve schwannomas, intracranial meningiomas, cataracts, and cutaneous tumors.
- NF1 results from mutation of the gene encoding neurofibromin.
- NF2 results from mutations of the gene encoding merlin.
- NF1 requires close follow-up to evaluate and manage the disease; patients have a decreased average lifespan.
- NF2 requires appropriate treatment of tumors and often leads to early death.

Review Questions

1. A 24-year-old male who initially presented with bilateral hearing loss and tinnitus demonstrates bilateral lesions within the internal acoustic canal on imaging. This patient most likely has a genetic defect affecting which chromosome?

- A. Chromosome 7
- B. Chromosome 11
- C. Chromosome 17
- D. Chromosome 21
- E. Chromosome 22

Explanation (requires correct answer)